

Solvency II & Regulatory Capital Implications

Session 16 • How GE-LAV changes insurance capital calculations

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Primary Text: Liquidity Illusion (Forthcoming, 2026)

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What we'll cover today

1

Solvency II overview

Pillar 1 quantitative requirements

2

Current PE treatment

Standard formula illiquidity premium

3

Why DCF marks mislead

Capital adequacy under stress

4

GE-LAV-based alternative

State-contingent risk weight

5

Industry implementation path

Realistic adoption timeline

Solvency II: how insurance regulators set PE capital

PILLAR 1

Quantitative requirements

Capital requirements (SCR), valuation rules, technical provisions. Private equity treated as 'Type 2 equity' with specific risk weight.

→ Standard formula: 49% capital charge

PILLAR 2

Governance & risk management

ORSA (Own Risk and Solvency Assessment) requires forward-looking risk analysis. Liquidity stress testing required.

→ Where GE-LAV most relevant

PILLAR 3

Disclosure

SFCR (Solvency and Financial Condition Report) and QRT (Quantitative Reporting Templates). Public disclosure of risk profile.

→ GE-LAV-informed disclosures

Current treatment: a flat 49% capital charge

Solvency II Standard Formula treats all PE holdings the same — 49% solvency capital regardless of state.

What this implies for capital allocation:

Scenario	Underlying NAV	Solvency Capital	Implied 'capital cost'
Normal markets (L=0)	\$100M reported	\$49M (49%)	≈ 4.9% per year of capital cost
Multi-shock (L=-0.5)	\$100M reported	\$49M (49%)	Same 4.9% — even though underlying is more risky
GFC depth (L=-1.5)	\$100M reported	\$49M (49%)	Same 4.9% — although true value is ~\$30M
Boom (L=+1)	\$100M reported	\$49M (49%)	Same 4.9% — although true value is ~\$115M

Flat 49% misallocates capital. Real risk varies dramatically with the liquidity state.

GE-LAV state-contingent capital charge for private equity

Y-axis: Capital charge (% of NAV)

X-axis: Liquidity state L

Legend:

- Current: flat 49%
- GE-LAV: state-contingent

Regime-specific charges:

- Boom ($L=+1$): 30%
- Normal ($L=0$): 49%
- Multi-shock ($L=-0.5$): 60%
- GFC ($L=-1.5$): 100%+

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Realistic adoption timeline

Solvency II rules change slowly. Realistic GE-LAV adoption path:

Phase 1 (2025-2026): Industry research

Trade associations (CFA Institute, IPEV) commission papers on stochastic liquidity premium

Light academic adoption

Phase 2 (2026-2028): ORSA stress testing

Pillar 2 (governance) requirement to incorporate GE-LAV into liquidity stress tests

Voluntary, internal use

Phase 3 (2028-2030): EIOPA review

European regulator considers standard-formula modification

Consultation period, no rule change yet

Phase 4 (2030-2032): Pillar 1 transition

Standard-formula recalibration to incorporate GE-LAV intuitions

May not be exact GE-LAV; will be GE-LAV-informed

Session 16 summary

What we accomplished today

- 1 Solvency II currently treats all PE with flat 49% capital charge — state-independent
- 2 This misallocates capital: too high in boom, too low in crisis
- 3 GE-LAV alternative: state-contingent charge ranging 30-100%+ based on $L(t)$
- 4 Realistic adoption timeline: 5-7 years through Pillar 2 first, Pillar 1 later

Next session

Session 17: Market clearing in the secondary market — GE-LAV's general equilibrium